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## Abstract

## 1 Introduction

## 2 Preliminaries

**Definition 2.1** (Positions and outcomes). A terminal position (no legal moves) is a P-position; a position is an N-position if and only if at least one legal move reaches a P-position; a position is a P-position if and only if every legal move reaches an N-position, the terminal case satisfying this clause vacuously.

**Definition 2.2** (The diagonal games  $D(m, r)$ ). Let  $m \geq 2$  and  $0 \leq r \leq m - 1$ .  $D(m, r)$  is played on the non-negative integers. From a position  $n \equiv r \pmod{m}$  (*squares mode*) a move subtracts any square  $k^2$  with  $k \geq 1$  and  $k^2 \leq n$ . From any other position (*fallback mode*) the only move subtracts  $m$ , available when  $n \geq m$ . Play is normal.

**Definition 2.3** (Squares modulo  $2m$ ).  $\mathcal{Q}(2m)$  denotes the set of residues of squares modulo  $2m$ . The mirror tool, proved once in general:  $(2m - k)^2 = 4m^2 - 4mk + k^2 \equiv k^2 \pmod{2m}$ , since  $4m^2 = 2m \cdot 2m$  and  $4mk = 2k \cdot 2m$ . So every table is  $k = 1..m$  plus mirrors, plus the 0 from  $k = 2m$ . Two side facts fell out of the mirrors and are used below: for odd  $m$ ,  $m^2 - m = m(m - 1)$  has the even factor  $m - 1$ , so  $m^2 \equiv m \pmod{2m}$ ; for even  $m$ ,  $m^2 = (m/2) \cdot 2m \equiv 0 \pmod{2m}$ .

| mod | $\mathcal{Q}(2m)$                          | complement                           |
|-----|--------------------------------------------|--------------------------------------|
| 4   | {0, 1}                                     | {2, 3}                               |
| 6   | {0, 1, 3, 4}                               | {2, 5}                               |
| 8   | {0, 1, 4}                                  | {2, 3, 5, 6, 7}                      |
| 10  | {0, 1, 4, 5, 6, 9}                         | {2, 3, 7, 8}                         |
| 12  | {0, 1, 4, 9}                               | {2, 3, 5, 6, 7, 8, 10, 11}           |
| 14  | {0, 1, 2, 4, 7, 8, 9, 11}                  | {3, 5, 6, 10, 12, 13}                |
| 18  | {0, 1, 4, 7, 9, 10, 13, 16}                | {2, 3, 5, 6, 8, 11, 12, 14, 15, 17}  |
| 22  | {0, 1, 3, 4, 5, 9, 11, 12, 14, 15, 16, 20} | {2, 6, 7, 8, 10, 13, 17, 18, 19, 21} |

Table 1: The engines at every modulus:  $\mathcal{Q}(2m)$  with complements.

### 3 The Chain Lemma

**Lemma 3.1** (Chain Lemma). *Fix integers  $m \geq 2$  and  $0 \leq r \leq m - 1$ , and let  $c$  be any residue with  $0 \leq c \leq m - 1$  and  $c \neq r$ . In  $D(m, r)$ , for every position  $n \geq 0$  with  $n \equiv c \pmod{m}$ :*

$$n \text{ is a P-position} \iff n \equiv c \pmod{2m}.$$

*Proof. Setup and notation.* Chain  $c$  is the set  $\{c + jm : j = 0, 1, 2, \dots\}$ , i.e. all positions  $\equiv c \pmod{m}$ ; call  $c + jm$  rung  $j$ . Two arithmetic remarks used throughout. (i) Every rung satisfies  $c + jm \equiv c \not\equiv r \pmod{m}$ , so every position of the chain is in fallback mode: its only candidate move is subtract- $m$ , legal exactly when the position is  $\geq m$ . (ii) Since  $n \equiv c \pmod{m}$ , the residue of  $n \bmod 2m$  is either  $c$  or  $c + m$ ; and  $n = c + jm \equiv c \pmod{2m} \iff 2m$  divides  $jm \iff j$  is even. So the Lemma is equivalent to: rung  $j$  is P iff  $j$  is even. The definition doing all the work: a terminal position is P; a position is N iff at least one legal move reaches a P-position; a position is P iff every legal move reaches an N-position, the terminal case being this clause satisfied vacuously.

**Closure.** Let  $n = c + jm$  be a rung. Its only candidate move is  $n \rightarrow n - m$ . If  $j = 0$ , the move is illegal, since  $n = c \leq m - 1 < m$ . If  $j \geq 1$ , the move is legal, since  $n = c + jm \geq 0 + m = m$ , and it lands on  $c + (j - 1)m$ : the same residue  $c \bmod m$ , hence rung  $j - 1$  of the same chain. So every legal move from a chain- $c$  position stays inside chain  $c$ , descending exactly one rung, and the set of positions reachable from the chain is contained in the chain.

**Independence.** Positions outside chain  $c$  — in particular class- $r$  positions firing squares — may well have moves that land on rungs of the chain. This cannot disturb the rungs' statuses: by the definition just quoted, the status of a position is a function only of the statuses of its options, the positions reachable from it; the definition never mentions moves arriving at a position. By Closure, every option of every rung is again a rung, so the statuses of chain  $c$  are computed entirely within chain  $c$  and would be identical if the rest of  $D(m, r)$  were deleted. Incoming traffic affects only the travellers, never the rungs.

**The bottom.** Rung 0 is the position  $c$ , the least element of the chain, with  $0 \leq c \leq m - 1$ . It is in fallback mode (because  $c \not\equiv r$ ) and its single candidate move is illegal (because  $c < m$ ). Both facts hold equally for  $c = 0$  and for  $c \geq 1$  — note that for  $c = 0$  no appeal to squares is needed:  $0 \neq r$  already puts the position 0 in fallback mode, where  $m \leq 0$  fails. So rung 0 is terminal, hence a P-position: “every legal move reaches an N” holds vacuously. This matches the claim, since  $j = 0$  is even.

**Alternation.** Claim: rung  $j$  is P for even  $j$  and N for odd  $j$ . Strong induction on  $j$ . Base,  $j = 0$ : done above. Step,  $j \geq 1$ : assume the claim for all smaller indices. By Closure, rung  $j$  has exactly one legal move — legal precisely because  $j \geq 1$ , the boundary check (rung 0 has no move at all; rung 1's move is legal since  $c + m \geq m$ ) — and it goes to rung  $j - 1$ . For a position with a single option  $x$ , the definition specializes: “some option is P” means  $x$  is P, so the position is N iff  $x$  is P; “every option is N” means  $x$  is N, so the position is P iff  $x$  is N. By the induction hypothesis, rung  $j - 1$  is P iff  $j - 1$  is even. Hence rung  $j$  is N iff  $j - 1$  is even, i.e. iff  $j$  is odd; and rung  $j$  is P iff  $j - 1$  is odd, i.e. iff  $j$  is even. Induction closed.

**Assembly.** Let  $c \not\equiv r$  be any class and  $n \equiv c \pmod{m}$  any position; write  $n = c + jm$  with  $j \geq 0$  uniquely determined. Closure and Independence license computing  $n$ 's status inside the chain alone; the bottom is the base of the induction and Alternation is its step, giving:  $n$  is P  $\iff j$  is even  $\iff jm \equiv 0 \pmod{2m} \iff n \equiv c \pmod{2m}$ . Nothing about  $c$  was used beyond  $0 \leq c \leq m - 1$  and  $c \neq r$ , so the result holds for all  $m - 1$  such classes simultaneously.  $\square$

## 4 Escape: the Clearing Lemma and its exceptions

The Lemma classifies no position of class  $r$  itself — it is silent about every  $n \equiv r \pmod{m}$ , and in particular does not assert those are N-positions; that claim is Milestone 2, and only the two together give the full mod- $2m$  law.

**Definition 4.1** (Window; free and hard residues; hard set). By the Chain Lemma (general form, as accepted): fallback moves preserve class mod  $m$ , so each class  $c \neq r$  is a single self-contained chain  $c, c + m, c + 2m, \dots$  with terminal bottom  $c$  — no square moves ( $c \not\equiv r$ ) and no affordable fallback ( $c < m$ ) — and P/N alternate upward from the stuck bottom: P exactly at  $n \equiv c \pmod{2m}$ . The window  $T(m, r)$  is the set of foreign bottoms mod  $2m$ . Class  $r$  splits mod  $2m$  into the **free residue** ( $r$  when  $r \geq 1$ , else  $m$ ) and the **hard residue** ( $r + m$  when  $r \geq 1$ , else  $0$ ). The free escape is immediate:  $1^2$  lands the free residue on the bottom directly beneath it ( $r - 1$ , or  $m - 1$  when  $r = 0$ ), and is legal from the smallest free member. For the hard residue, a square value  $s \pmod{2m}$  lands in the window iff

$$s \in H(m, r) = \{r+1, \dots, m-1\} \cup \{m+1, \dots, m+r\} \text{ for } r \geq 1, \quad \text{or} \quad \{m+1, \dots, 2m-1\} \text{ for } r = 0.$$

**Lemma 4.2** (Constitutional uselessness). *For every  $m$  and  $r$ :  $0, 1, m \notin H(m, r)$ ; a square  $\equiv 0$  or  $m \pmod{2m}$  is  $\equiv 0 \pmod{m}$  and keeps a class- $r$  position inside class  $r$ ; and subtracting a square  $\equiv 1 \pmod{2m}$  from the hard residue lands one below it in the top half, an N-position.*

*Proof.* Constitutional uselessness, for every  $m$  and  $r$ :  $0, 1, m \notin H(m, r)$  — a square  $\equiv 0$  or  $m \pmod{2m}$  is  $\equiv 0 \pmod{m}$  and stays inside class  $r$ , and subtracting  $\equiv 1$  from the hard residue lands one below it in the top half, an odd rung, N.  $\square$

**Lemma 4.3** (Clearing Lemma). *Suppose some square  $s \neq m$  lies in the integer interval  $(r, m+r]$  when  $r \geq 1$ , or in  $(m, 2m)$  when  $r = 0$ . Then the entire hard class is N by the single move  $-s$ .*

*Proof.* (Formerly the Zone Lemma — the parity audit showed it is not a patch on the odd- $m$  witness but the escape engine itself, parity-free.) Legality:  $s \leq m + r$ , the smallest hard member (for  $r = 0$ :  $s < 2m$ , and the smallest positive hard member is  $2m$ ), so the move is affordable to every member from the first. Landing:  $n - s \equiv m + r - s \pmod{2m}$  is a foreign bottom — in  $(r, m)$  when  $r < s < m$ , in  $[0, r)$  when  $m < s \leq m + r$ ; for  $r = 0$ , the landing  $2m - s$  lies in  $(0, m)$ . Combined with the free escape, class  $r$  is all N — except possibly  $n = 0$ , handled in Lemma 4.7.  $\square$

**Lemma 4.4** (The interval contains a square). *For  $m \geq 3$  and  $1 \leq r \leq m - 1$ , the interval  $(r, m + r]$  contains a square. For  $r = 0$ , the interval  $(m, 2m)$  contains a square for every  $m \geq 3$  except  $m = 4$ .*

*Proof.* (IV.3: the interval always contains a square ( $m \geq 3$ ) — repaired and parity-free.) For  $r \geq 1$ : if  $(r, m + r]$  contained no square, consecutive squares would straddle it,  $c^2 \leq r$  and  $(c + 1)^2 > m + r$ , so  $2c + 1 > m$ , hence  $c \geq m/2$  for any  $m$ , hence  $c^2 \geq m^2/4 > m - 1 \geq r$ , since  $m^2/4 - (m - 1) = (m - 2)^2/4 > 0$  for  $m \geq 3$  — contradicting  $c^2 \leq r$ . The inequality degenerates at exactly one modulus,  $m = 2$ , where  $(m - 2)^2 = 0$  and the interval can indeed be empty ( $(1, 3]$  in  $D(2, 1)$  contains no square). Erratum, folded in: for  $r = 0$  the interval reads  $(m, 2m)$ , which contains a square for every  $m \geq 3$  **except**  $m = 4$  —  $(4, 8) = \{5, 6, 7\}$  is square-free, and it is the lone counterexample, since the same consecutive-squares squeeze forces emptiness only for  $m \leq 5$ , and  $m = 3$  has its 4,  $m = 5$  its 9.  $\square$

**Proposition 4.5** (The failure window). *The hypothesis of Lemma 4.3 fails for  $r \geq 1$  only if  $m = b^2$  with  $(b-1)^2 \leq r \leq 2b$ , which is possible only for  $b \leq 3$ : among  $m \geq 3$ , exactly  $m = 4$  ( $r \in \{1, 2, 3\}$ ) and  $m = 9$  ( $r \in \{4, 5, 6\}$ ).*

*Proof.* With a square always present, the Clearing Lemma's hypothesis can fail for  $r \geq 1$  only if the *unique* square in  $(r, m+r]$  is  $m$  itself:  $m = b^2$  with  $(b-1)^2 \leq r$  (no smaller square intrudes) and  $(b+1)^2 > m+r$ , i.e.  $r \leq 2b$ . The window  $(b-1)^2 \leq r \leq 2b$  is nonempty iff  $(b-1)^2 \leq 2b$  iff  $b \leq 3$ . Among  $m \geq 3$  that leaves exactly  $m = 4$  ( $b = 2$ ,  $r \in \{1, 2, 3\}$ ) and  $m = 9$  ( $b = 3$ ,  $r \in \{4, 5, 6\}$ ). With the  $r = 0$  erratum, every class of  $m = 4$  is condemned from the outside — no clearing square exists for any  $r$  — while  $m = 9$  is healthy at every  $r$  except 4, 5, 6. Squarehood of  $m$  alone is harmless: at  $m = 25$  ( $b = 5$ ) the neighboring square 36 sits only  $2b+1 = 11$  above  $m$ , deep inside a 25-wide window, so it invades  $(r, m+r]$  for every  $r$ ; a square  $m$  isolates itself only when the modulus is small enough ( $b \leq 3$ ) for square-gaps to outrun  $m$ .  $\square$

**Proposition 4.6** (The rebels). *For  $(m, r) \in \{(9, 4), (9, 5), (9, 6)\}$  the position  $n = m + r$  is a P-position, and it is the unique P-position of  $D(m, r)$  outside the window.*

*Proof.* At  $(9, r)$  with  $r \in \{4, 5, 6\}$ , the smallest hard member  $n_0 = m + r \in \{13, 14, 15\}$  can afford only the squares  $k^2 \leq r$  and 9. A square  $k^2 \leq r$  lands on  $m + (r - k^2)$ , the odd rung directly above bottom  $r - k^2$  — N. The square 9 lands on  $r$  itself, the smallest free member — N, since it escapes by  $1^2$ . Every move lands on N, so  $n_0$  is P. In-game receipts: in  $D(9, 4)$ ,  $13 \rightarrow 12, 9, 4$ , all N; in  $D(9, 5)$ ,  $14 \rightarrow 13, 10, 5$ , all N; in  $D(9, 6)$ ,  $15 \rightarrow 14, 11, 6$ , all N. Above the rebel the class closes uniformly: the members  $27 + r$  (31, 32, 33) are cleared by  $25 \equiv 7 \pmod{18}$ , which lies in all three hard sets, landing on bottoms 6, 7, 8; the members  $45 + r$  (49, 50, 51) by  $49 \equiv 13$ , landing on bottoms 0, 1, 2; and from  $63 + r \geq 64 = (m-1)^2$  onward, the witness  $64 \equiv 10 = m + 1$  lands on bottom  $r - 1$  forever. So each exceptional game carries exactly one extra P-position, and the exception set is precisely the  $E$  displayed in the box.  $\square$

**Lemma 4.7** (Boundary positions corrupt nothing). *Corruption audits.* *The  $n = 0$  clause: when  $r = 0$ , zero is claimed by squares mode and is the unique squares-mode position that can afford no square ( $k \geq 1$  forces  $k^2 \geq 1 > 0$ ), so it is terminal, hence P — while the unamended law would file residue 0 as N. It corrupts nobody: fallback chains never visit class 0; the positions that can drop to 0 are  $n = k^2 \equiv 0 \pmod{m}$ , class-0 members the theorem already declares N; and the classes above escape without it — spot receipts:  $10 - 9 = 1$  (P) in  $D(5, 0)$ ,  $14 - 9 = 5$  (P) in  $D(7, 0)$ . For  $r \geq 1$  the clause is absorbed, since 0 is itself a foreign bottom; it earns its keep exactly at  $r = 0$ . The rebels, one sentence in the same mirror: a rebel corrupts nobody either — fallback chains never enter class  $r$ , and the only positions with a move onto  $m + r$  are class- $r$  members above it (a square step back into class  $r$  forces  $k^2 \equiv 0 \pmod{9}$ , i.e.  $3 \mid k$ ), which the theorem already declares N, so the rebel's P-ness merely hands them one more witness for what they already were.*

## 5 The Diagonal Theorem

**Theorem 5.1** (Diagonal Law). *Let  $m \geq 3$  with  $m \neq 4$ , and let  $0 \leq r \leq m - 1$ . Define  $E = \{(9, 4), (9, 5), (9, 6)\}$ . In  $D(m, r)$ , a position  $n$  is a P-position if and only if*

$$n \equiv b \pmod{2m} \text{ for some foreign bottom } b \in \{0, 1, \dots, m-1\} \setminus \{r\}, \quad \text{or} \quad n = 0, \quad \text{or} \quad (m, r) \in E \text{ and } n = m + r$$

*The law holds for both parities of  $m$ , with no thresholds and no unverified zones.*

WRITE: assembly paragraph. □

*Remark 5.2* (The historical witness). *The odd- $m$  witness identity, the original route.* For odd  $m$ ,  $(m-1)^2 = m^2 - 2m + 1 \equiv m^2 + 1 \equiv \mathbf{m} + \mathbf{1} \pmod{2m}$ , using  $m^2 \equiv m$  from Part III; and  $m+1$  belongs to every hard set, landing the hard residue exactly on bottom  $r-1$  ( $m-1$  when  $r=0$ ). So the pair  $(1^2, (m-1)^2)$  is a universal witness pair for all odd  $m$  at once — at the price of the threshold  $n \geq (m-1)^2$ . The Clearing Lemma supersedes it: a square below  $m+r$  does the same job with legality free, for both parities, with no threshold. The identity remains the cleanest line in this file and the historical door to everything above.

## 6 The diseased moduli

**Lemma 6.1** (Confinement Lemma). *For  $m \in \{2, 4\}$  and every  $r$ :  $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$ , while  $H(m, r)$  excludes 0, 1, and  $m$ . Hence no square move from a hard-residue position reaches the window: every move stays inside class  $r$  or lands on an N-position of a foreign chain.*

*Proof.* For  $m \in \{2, 4\}$  and every  $r$ :  $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$  — read directly off the mod-4 and mod-8 rows of Table 1 — while every hard set  $H(m, r)$  excludes 0, 1, and  $m$  (Definition 4.1). Hence from any hard-residue position, every legal square move either stays inside class  $r$  (a square  $\equiv 0$  or  $m \pmod{2m}$  is  $\equiv 0 \pmod{m}$ ) or lands one below on the odd rung above a bottom (a square  $\equiv 1$ ), an N-position — never on a P-position of a foreign chain. *Proof:* only-options enumeration over  $\mathcal{Q}(2m)$ , the mirror image of the Independence section — there, fallback moves cannot leave a chain; here, square moves cannot leave the class except onto N. *Consequence:* the hard class's P/N structure is decided by a self-contained internal recursion, and its very first member already defies the chain law: 3 is P in  $D(2, 1)$  and 5 is P in  $D(4, 1)$ , where escape, if it existed, would force N. Calibration, as accepted at the viva: for  $m=2$  and  $m=4$  we prove confinement — no square door into the window, so the class turns inward. We do **not** prove that these confined classes obey no eventual law of their own: “no period and no modulus up to 200 fits” is a reading from the search instrument, not a theorem, and their aperiodicity remains open. □

*Remark 6.2* (Double condemnation). The two diseased moduli are exactly the two the parity-free engine cannot serve. At  $m=2$  the engine is silent: the always-a-square inequality degenerates  $((m-2)^2 = 0)$  and the clearing interval  $(1, 3]$  is genuinely square-free. At  $m=4$  the engine convicts every class from the outside: the failure window covers  $r \in \{1, 2, 3\}$ , and the  $r=0$  interval  $(4, 8)$  is square-free. Outside view and inside view agree because both are reading the same two table rows:  $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$ .

## 7 Further solved games

**Example 7.1** ( $D(5, 2)$ , solved). A position  $n$  of  $D(5, 2)$  is a P-position if and only if  $n \equiv 0, 1, 3$ , or  $4 \pmod{10}$ .

**II.1 Statement.** There is an explicit threshold  $N_0$ , computed in II.6 below, such that every position  $n \equiv 2 \pmod{5}$  with  $n \geq N_0$  has at least one legal square move — some  $k$  with  $k^2 < n$  — whose landing  $n - k^2$  is a P-position of a fallback chain. Consequently every such  $n$  is an N-position.

**II.2 The target window.** By the Chain Lemma, the fallback classes  $0, 1, 3, 4 \pmod{5}$  have their P-positions exactly at  $n \equiv 0, 1, 3, 4 \pmod{10}$ ; call that set  $T = \{0, 1, 3, 4\} \pmod{10}$ . A

landing in  $T$  is genuinely P: a number  $\equiv 0, 1, 3$ , or  $4 \pmod{10}$  is  $\equiv 0, 1, 3$ , or  $4 \pmod{5}$ , so it lives in a fallback chain where the lemma has jurisdiction, and its residue mod 10 matching the chain bottom's means it sits an even number of rungs up — precisely the lemma's P-rungs.

**II.3 The split.** Write  $n = 5q + 2$ : if  $q = 2t$  then  $n = 10t + 2$ , and if  $q = 2t + 1$  then  $n = 10t + 7$ , so a class-2 position is  $\equiv 2$  or  $7 \pmod{10}$  — that is,  $r$  or  $r + m$  — and the proof forks on the parity of  $q$  exactly the way the chain proof forked on the parity of  $j$ .

**II.4 Coverage line one.** For  $n \equiv 2 \pmod{10}$ : the differences  $2 - T$  are  $\{2 - 0, 2 - 1, 2 - 3, 2 - 4\} \equiv \{2, 1, 9, 8\} \pmod{10}$ , and intersecting with  $\mathcal{Q} = \{0, 1, 4, 5, 6, 9\}$  leaves  $\{1, 9\}$ ; take the smallest,  $s = 1$ , so subtracting  $1^2$  gives  $n - 1 \equiv 1 \pmod{10} \in T$ .

**II.5 Coverage line two.** For  $n \equiv 7 \pmod{10}$ : the differences  $7 - T$  are  $\{7, 6, 4, 3\} \pmod{10}$ , intersecting with  $\mathcal{Q}$  leaves  $\{4, 6\}$ ; take  $s = 4$ , so subtracting  $2^2$  gives  $n - 4 \equiv 3 \pmod{10} \in T$ . Both keystones came with a spare —  $9 = 3^2$  also rescues residue 2, and  $16 = 4^2$  also rescues residue 7. The corridor is wider than it needs to be.

**II.6 Legality and the threshold.** The rescue squares are the concrete numbers 1 and 4. For the residue-2 class, whose members are  $2, 12, 22, \dots$ , legality demands  $1 \leq n$ , which already holds at the smallest member. For the residue-7 class, members  $7, 17, 27, \dots$ , legality demands  $4 \leq n$ , which holds at 7. So both rescues are legal for every member of their class, and the threshold collapses:  $\mathbf{N}_0 = \mathbf{2}$ , the smallest class-2 position in existence. Below  $N_0$  there are no class-2 positions at all, so the hand-check zone is empty — verdict: vacuous. Checking the workbook rows anyway, as ordered:  $2 - 1 = 1$  (P) ✓,  $7 - 4 = 3$  (P) ✓,  $12 - 1 = 11$  (P) ✓,  $17 - 4 = 13$  (P) ✓,  $22 - 1 = 21$  (P) ✓,  $27 - 4 = 23$  (P) ✓. Every row is rescued-anyway, rebels: none. Worth recording why it collapsed: the rescue squares 1 and 4 happen to be smaller than the smallest members of their classes. Nothing guarantees that luck at other moduli, which is exactly why the theorem was written with  $N_0$  as a named unknown.

**II.7 Assembly.** The Chain Lemma classifies the four fallback classes everywhere: P exactly at  $n \equiv 0, 1, 3, 4 \pmod{10}$ , N at  $5, 6, 8, 9$ . The Escape Theorem classifies class 2 everywhere, since  $N_0 = 2$  means “past the threshold” is all of it: residues 2 and  $7 \pmod{10}$  are all N. Union the two and every residue mod 10 is accounted for, with no unexplained positions beneath any threshold.

**Theorem ( $D(5, 2)$ , solved).** A position  $n$  is P if and only if  $n \equiv 0, 1, 3$ , or  $4 \pmod{10}$ .

One coincidence too pretty not to flag: the P-set of the whole game *is* the target window  $T$  — the landing pads and the answer turned out to be the same four residues. [Resolved: Remark 7.3.]

**II.8 Scope and the forward hook.** The proof secretly depended on squares mod 10 being *rich*:  $\mathcal{Q}$  fills six of the ten residues, enough that both difference lists  $\{2, 1, 9, 8\}$  and  $\{7, 6, 4, 3\}$  caught at least one member — had  $\mathcal{Q}$  missed either list entirely, one residue of class 2 would have had no escape and the whole corridor would have collapsed. Generalizing to  $D(m, r)$  therefore means asking whether that richness survives at other moduli — whether squares mod  $2m$  keep intersecting the difference sets  $\rho - T$  for both  $\rho \in \{r, r + m\}$  — and since quadratic residues thin out and organize themselves as the modulus grows, that question, not the game logic, is the real content of what follows.

*Remark 7.2* (Verification against the workbook). Prediction first, then the rows: residue-2 positions  $(2, 12, 22)$  should be rescued by any legal square  $\equiv 1$  or  $9 \pmod{10}$ , residue-7 positions  $(7, 17)$  by any square  $\equiv 4$  or  $6 \pmod{10}$ , and every other square should be wasted.

- $n = 2$ : only move  $2 - 1 = 1$  (P). Predicted rescue  $1^2$  present ✓, nothing else in the row to waste.
- $n = 7$ :  $7 - 1 = 6$  (N),  $7 - 4 = 3$  (P). Rescue  $2^2$  present ✓; the  $1^2$  is wasted exactly as the congruence says ( $1 \notin \{4, 6\} \pmod{10}$ , landing  $\equiv 6$ ).
- $n = 12$ :  $12 - 1 = 11$  (P),  $12 - 4 = 8$  (N),  $12 - 9 = 3$  (P). Primary rescue  $1^2$  ✓, and the spare  $9 = 3^2$  makes its first appearance in the first row big enough to afford it.
- $n = 17$ :  $17 - 1 = 16$  (N),  $17 - 4 = 13$  (P),  $17 - 9 = 8$  (N),  $17 - 16 = 1$  (P). Primary  $2^2$  ✓ and spare  $4^2$  ✓.
- $n = 22$ :  $22 - 1 = 21$  (P),  $22 - 4 = 18$  (N),  $22 - 9 = 13$  (P),  $22 - 16 = 6$  (N). Primary  $1^2$  ✓, spare  $3^2$  ✓.

The check came back stronger than required: not only does the predicted rescue sit in every row, every wasted square is wasted for the predicted reason — its residue misses the survivor set, and every wasted landing is  $\equiv 6$  or  $8 \pmod{10}$ , both N-residues. The congruence machinery reproduces the workbook letter-for-letter, and the 200,000 mechanical sweep stands as the bulk consistency evidence behind it.

*Remark 7.3* (The  $D(5, 2)$  coincidence, resolved). For any escaped game the P-set equals its window by construction:  $\text{P-set} = (\text{chain P's}) \cup (\text{class-}r \text{ P's}) = T \cup \emptyset = T$ . The equality fails exactly where escape does — the three exceptional games have  $\text{P-set} = T \cup \{m + r\}$ , and the diseased games' P-sets properly contain their windows.

**Theorem 7.4** (Foursquare). *Consider the game on  $n \geq 0$ : if  $n \equiv 1 \pmod{4}$ , subtract any positive square  $\leq n$ ; otherwise subtract  $s \in \{3, 8, 9\}$ ,  $s \leq n$ ; normal play. Then  $n$  is a P-position if and only if  $n \equiv 0, 2, 4$ , or  $6 \pmod{16}$ , for all  $n \geq 0$  with no preperiod.*

*Partition Exhaustiveness*: The set of residues modulo 16 is exhaustively partitioned into claimed P-positions  $\mathcal{P} = \{0, 2, 4, 6\} \pmod{16}$  and claimed N-positions  $\mathcal{N} = \{1, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15\} \pmod{16}$ , such that  $\mathcal{P} \cup \mathcal{N} = \mathbb{Z}/16\mathbb{Z}$  and  $\mathcal{P} \cap \mathcal{N} = \emptyset$ .

**Lemma 7.5** (Orientation Lemma). *Every state  $n \equiv 0, 2, 4$ , or  $6 \pmod{16}$  is even. Consequently,  $n \not\equiv 1 \pmod{4}$ .*

*Proof.* For any  $r \in \{0, 2, 4, 6\}$ ,  $n = 16q + r \equiv r \equiv 0 \pmod{2}$ . Since  $n$  is even,  $n \pmod{4} \in \{0, 2\}$ , which implies  $n \not\equiv 1 \pmod{4}$ . Thus, Square Mode never applies at any claimed P-position, and for  $n \geq 9$ , the full move set  $\{3, 8, 9\}$  is strictly available.  $\square$

**Lemma 7.6** (Odd Squares Modulo 16 Lemma). *For any odd integer  $k \in \mathbb{Z}$ ,  $k^2 \equiv 1$  or  $9 \pmod{16}$ .*

*Proof.* Express any odd integer as  $k = 2m + 1$  for some  $m \in \mathbb{Z}$ . Squaring yields:

$$k^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 4m(m + 1) + 1$$

Because  $m(m + 1)$  is the product of two consecutive integers, it is always even. Let  $m(m + 1) = 2j$  for  $j \in \mathbb{Z}$ . Substituting gives:

$$k^2 = 4(2j) + 1 = 8j + 1$$

If  $j$  is even ( $j = 2l$ ),  $k^2 = 16l + 1 \equiv 1 \pmod{16}$ . If  $j$  is odd ( $j = 2l + 1$ ),  $k^2 = 16l + 9 \equiv 9 \pmod{16}$ . Thus, every odd square is congruent to 1 or 9  $\pmod{16}$ .  $\square$

*Proof of Theorem 7.4.* (By Strong Induction on  $n$ .)

**Base Cases** ( $0 \leq n \leq 8$ ). Direct manual calculation of legal move bounds and reachable states confirms:

- $n = 0$ : Standard mode, no moves  $\leq 0$ . **P-position.**
- $n = 1$ : Square mode, move  $s = 1^2 \rightarrow 0$  (P). **N-position.**
- $n = 2$ : Standard mode, no moves  $\leq 2$ . **P-position.**
- $n = 3$ : Standard mode, move  $s = 3 \rightarrow 0$  (P). **N-position.**
- $n = 4$ : Standard mode, move  $s = 3 \rightarrow 1$  (N). All legal moves go to N. **P-position.**
- $n = 5$ : Square mode, moves  $s = 1^2 \rightarrow 4$  (P),  $s = 2^2 \rightarrow 1$  (N). **N-position.**
- $n = 6$ : Standard mode, move  $s = 3 \rightarrow 3$  (N). **P-position.**
- $n = 7$ : Standard mode, move  $s = 3 \rightarrow 4$  (P). **N-position.**
- $n = 8$ : Standard mode, move  $s = 8 \rightarrow 0$  (P). **N-position.**

**Inductive Step** ( $n \geq 9$ ). Assume the theorem holds for all states  $m < n$ .

*Part 1: Closure (Claimed P-positions).* Let  $n \equiv r \pmod{16}$  with  $r \in \{0, 2, 4, 6\}$ . By Lemma 7.5,  $n \not\equiv 1 \pmod{4}$ . For all  $n \geq 9$ , all of  $\{3, 8, 9\}$  are legal moves since  $s \leq n$ . We compute  $r - s \pmod{16}$  across all 12 combinations:

- $r = 0 \implies 0 - 3 \equiv 13, \quad 0 - 8 \equiv 8, \quad 0 - 9 \equiv 7 \pmod{16}$
- $r = 2 \implies 2 - 3 \equiv 15, \quad 2 - 8 \equiv 10, \quad 2 - 9 \equiv 9 \pmod{16}$
- $r = 4 \implies 4 - 3 \equiv 1, \quad 4 - 8 \equiv 12, \quad 4 - 9 \equiv 11 \pmod{16}$
- $r = 6 \implies 6 - 3 \equiv 3, \quad 6 - 8 \equiv 14, \quad 6 - 9 \equiv 13 \pmod{16}$

None of the resulting residues belong to  $\{0, 2, 4, 6\} \pmod{16}$ . By the strong induction hypothesis, every reachable state  $n - s < n$  is an N-position. Hence,  $n$  is a **P-position**.

*Part 2: Escape (Claimed N-positions).* Let  $n \equiv r \pmod{16}$  with  $r \in \mathcal{N}$ . We display a legal move  $s$  such that  $n - s \in \mathcal{P}$ :

1. **Non-Square Residues** ( $r \in \{3, 7, 8, 10, 11, 12, 14, 15\}$ ):

- $r = 3$ :  $s = 3 \implies 3 - 3 \equiv 0 \in \mathcal{P}$
- $r = 7$ :  $s = 3 \implies 7 - 3 \equiv 4 \in \mathcal{P}$
- $r = 8$ :  $s = 8 \implies 8 - 8 \equiv 0 \in \mathcal{P}$
- $r = 10$ :  $s = 8 \implies 10 - 8 \equiv 2 \in \mathcal{P}$
- $r = 11$ :  $s = 9 \implies 11 - 9 \equiv 2 \in \mathcal{P}$
- $r = 12$ :  $s = 8 \implies 12 - 8 \equiv 4 \in \mathcal{P}$
- $r = 14$ :  $s = 8 \implies 14 - 8 \equiv 6 \in \mathcal{P}$
- $r = 15$ :  $s = 9 \implies 15 - 9 \equiv 6 \in \mathcal{P}$

Since  $n \geq 9$ ,  $s \leq n$  is satisfied in all cases.

2. **Square-Mode Residues** ( $r \in \{1, 5, 9, 13\}$ ): By Lemma 7.6, odd squares satisfy  $k^2 \equiv 1$  or  $9 \pmod{16}$ . Using  $s = 1^2 = 1$  or  $s = 3^2 = 9$ :

- $r = 1$ :  $s = 1 \implies 1 - 1 \equiv 0 \in \mathcal{P}$
- $r = 5$ :  $s = 1 \implies 5 - 1 \equiv 4 \in \mathcal{P}$
- $r = 9$ :  $s = 9 \implies 9 - 9 \equiv 0 \in \mathcal{P}$
- $r = 13$ :  $s = 9 \implies 13 - 9 \equiv 4 \in \mathcal{P}$

For  $r \in \{1, 5\}$ ,  $s = 1 \leq n$  holds for  $n \geq 1$ . For  $r \in \{9, 13\}$ ,  $s = 9 \leq n$  holds for  $n \geq 9$ .

By strong induction, every state  $n \in \mathcal{N}$  can transition to a P-position  $n - s < n$ . Thus,  $n$  is an **N-position**.  $\square$

*Remark 7.7 (No-Preperiod Remark).* The periodicity of P-positions (period = 16) holds strictly and purely from  $n = 0$  onward, exhibiting no initial non-periodic transient phase (preperiod = 0).



**Proposition 7.8** (Odd-fallback collapse at  $m = 2$ ). (Lemma A.) *For every odd integer  $a \geq 1$ , the set of P-positions (states  $n$  with  $G(n) = 0$ ) consists precisely of all even integers. Consequently, the emergent period is 2 with a residue-set size of 1.*

(Lemma B.) *For all odd fallbacks  $a, a' \geq 1$  and all states  $n \geq 0$ , the Grundy values satisfy  $G_a(n) = G_{a'}(n)$ .*

*Proof. Setup.* We fix  $r = 1$ : squares fire on odd  $n$ , and the fallback rule applies on even  $n$ , making the allowed move from  $n$  to  $n - a$ . Let  $G(n)$  denote the Grundy value of state  $n$ , with  $G(0) = 0$ .

*Lemma A (P-positions = Even Numbers).* We proceed by induction on  $n$ .

**Base case** ( $n = 0$ ). Terminal position,  $G(0) = 0$ , so 0 is a P-position.

**Inductive step.** Assume for all  $k < n$  that  $G(k) = 0 \iff k$  is even (equivalently,  $G(k) > 0 \iff k$  is odd).

**Case 1:  $n$  is even.** The only available move is subtraction of the odd fallback  $a$ :  $n \rightarrow n - a$ . Since  $n$  is even and  $a$  is odd, the destination  $n - a$  is strictly odd. By the induction hypothesis,  $G(n - a) \neq 0$  (if  $n - a \geq 0$ ), or the move set is empty (if  $a > n$ ).

- If  $a > n$ : the move set is  $\emptyset$ , so  $\text{mex}(\emptyset) = 0$  and  $G(n) = 0$ .
- If  $a \leq n$ : the set of reachable values is  $\{G(n - a)\}$  with  $G(n - a) \neq 0$ , so  $\text{mex}(\{G(n - a)\}) = 0$  and  $G(n) = 0$ .

In both cases  $G(n) = 0$ . Thus all even positions are P-positions.

**Case 2:  $n$  is odd.** The move set consists of square subtractions  $n \rightarrow n - k^2$  for all integers  $k \geq 1$  with  $k^2 \leq n$ . Choosing  $k = 1$  gives a legal move to  $n - 1$ . Since  $n$  is odd,  $n - 1$  is even, so  $G(n - 1) = 0$  by the strong induction hypothesis. Thus  $0 \in \{G(n - k^2) : k^2 \leq n\}$ , forcing the mex to be positive, so  $G(n) \neq 0$ .

Therefore  $G(n) = 0 \iff n \equiv 0 \pmod{2}$ .

*Lemma B (Full Sequence Equivalence).* **Answers to the Referee Questions.**

**Question 1 (even positions).** What is  $\text{mex}(\{v\})$  when  $v \neq 0$ , and what does that force  $G(n)$  to be for even  $n$ ?

For even  $n$ , the reachable set is either empty (if  $n < a$ ) or  $\{G(n - a)\}$ . By Lemma A, since  $n - a$  is odd,  $G(n - a) = v > 0$ . Now  $\text{mex}(\emptyset) = 0$ , and  $\text{mex}(\{v\}) = 0$  whenever  $v > 0$ . Therefore  $G(n) = 0$  for every even  $n$ , regardless of the specific value of  $v$ , and regardless of whether the move exists at all. The parameter  $a$  evaporates at every even step.

**Question 2 (odd positions).** At odd  $n$ , split the available squares  $k^2$  by parity: which earlier positions' Grundy values enter the mex?

For odd  $n$ : if  $k$  is even,  $k^2$  is even, so  $n - k^2$  is odd; if  $k$  is odd,  $k^2$  is odd, so  $n - k^2$  is even. By Lemma A, for every odd  $k$  the landing  $n - k^2$  is even with  $G(n - k^2) = 0$ . Thus the reachable Grundy values from odd  $n$  are  $\{0\} \cup \{G(n - k^2) : k \text{ even}, k^2 \leq n\}$ . The value 0 is guaranteed by the odd square moves ( $k = 1, 3, 5, \dots$ ), and the non-zero entries depend exclusively on  $G(n - k^2)$  for even  $k$  — landings on smaller odd numbers. So  $G$  at odd  $n$  is computed purely from earlier odd states, completely bypassing all even positions, and thus bypassing  $a$  entirely.

By induction on  $n$ :  $G(0) = 0$ ; every even state yields  $G = 0$  independent of  $a$ ; and every odd state's value is computed from prior odd states via even-square subtractions, in which  $a$  nowhere appears. The parameter  $a$  never impacts the Grundy sequence, so all odd fallbacks collapse to the exact same game.  $\square$

## 8 Discovery methodology and retrodictions

## 9 Open question

## Acknowledgments and AI disclosure

## References

- [1] E. Duchêne and M. Rigo, Invariant games, *Theoret. Comput. Sci.* 411 (2010) 3169–3180.
- [2] S. W. Golomb, A mathematical investigation of games of “take-away”, *J. Combin. Theory* 1 (1966) 443–458. [check]
- [3] U. Larsson and I. Saha, A brief conversation about subtraction games, arXiv:2405.20054 (2024).
- [4] A. Siegel, Finite excluded subtraction sets and infinite modular Nim, M.Sc. thesis, Dalhousie University, 2005. [check]
- [5] D. Sleator and M. Slusky, Subtraction games with FES sets of size 3, arXiv:1201.3299 (2012).
- [6] A. Holshouser and H. Reiter, One pile Nim with arbitrary move function, *Electron. J. Combin.* 10 (2003), #N7.
- [7] N. Fox, On aperiodic subtraction games with bounded Nim sequence, arXiv:1407.2823 (2014).
- [8] R. J. Nowakowski, Unsolved problems in combinatorial games, in *Games of No Chance 6* (U. Larsson, ed.), Math. Sci. Res. Inst. Publ. 71, Cambridge University Press, 2025, p. 43.